

ASSIGNMENT #02

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, February 13th TA's Mailbox

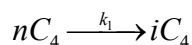
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Plant Wide Process Control.

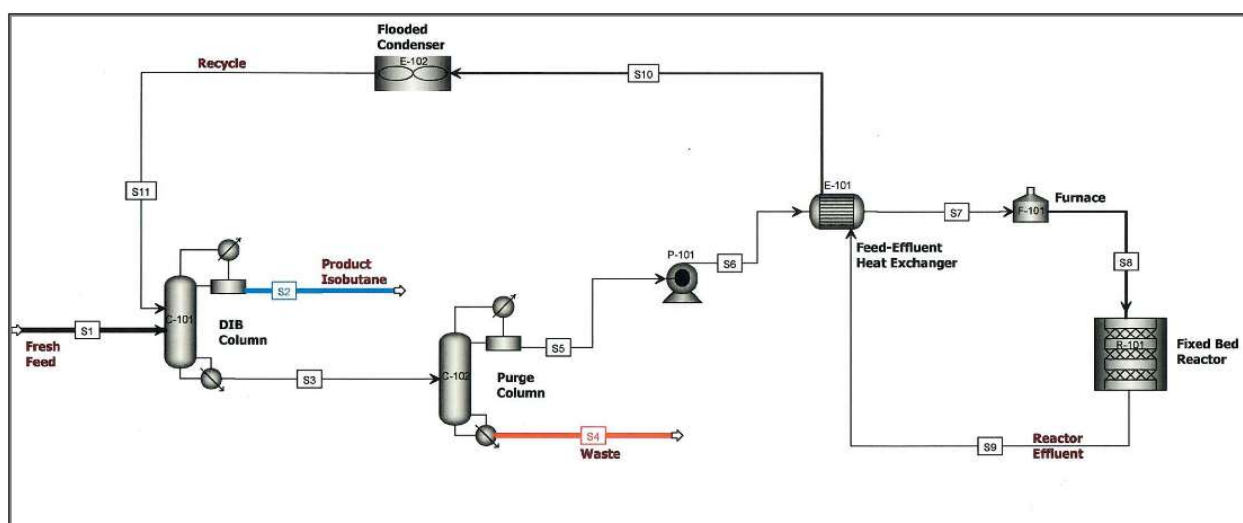
A process flowsheet diagram for the production of iso-butane via the isomerization of n-butane is shown in the Figure below: The isomerization reaction takes place in the fixed bed reactor with pressure drop equal to 15 psi. The kinetics of the reaction are shown below:



$$\mathcal{R}_1 = k_{10} \exp\left\{-\frac{E}{RT}\right\} C_{nC_4}$$

$$E = 30,000 \text{ Btu} / \text{lbmol}$$

$$k_{10} = 4 \times 10^8 \text{ l} / \text{hr}$$



To better understand the process, open the attached ASPEN simulation and carefully study the operating conditions of all unit operations. The distillate product from the DeIsoButanizer column (DIB) has a specification of 2 mol% n-butane and is directed to a storage tank. The plant uses cooling water from near-by cooling tower designed with 75% efficiency with hot water inlet temperature of 120 °F and air ambient temperature of 90 °F with relative humidity 70%. A steam plant from a nearby facility supplies saturated steam (with quality 0.95) at three different levels of pressure: low pressure at 50 psig, medium pressure at 150 psig and high pressure at 600 psig. Natural gas from an existing pipe line is used for the fired heater that heats the feed to the reactor. The heat transfer coefficients in all the heat exchangers can be taken as the middle values of the ranges that are listed in the **CHE Calculator**.

Assume that the pressure drop in the heat exchanger is 10 psi both on the tube side and 5 psi on the shell side and the pressure drop in the fired heater is 60 psi. The pressure drop in the liquid line from the pump discharge to the inlet of the fired heater is approximately 8 psi.

This assignment has three parts:

Part A: Flowsheet Modifications

1. First you must modify the simulation to obtain the product specifications and the diameters of the distillation columns. That will help you calculate the height of the liquid in the column sump. Some modifications will also be need to accommodate the additional information on pressure drop given in the problem description above.

Using ASPEN Plus, the simulation was run and the equipment sizing was performed using the Economics module of the software. The following table shows the design generated design specifications.

Item description	C-101-	C-102-
User tag number	TW	TW1
Remarks 1	Equipment mapped	Equipment mapped
Quoted cost per item		
Currency unit for matl cost		
Number of identical items		
Installation option		
Application		
Base material Bottom		
Diameter Bottom section [ft]	1.5	0.5
Bottom tangent to tangent height [ft]	156	70

Figure 1 Distillation Column Design Parameters

2. Second you must create three utilities (cooling water, steam and natural gas) and attach them to the appropriate unit operations. Utility consumption will be used to calculate the steady-state values of the manipulated variables for the process control loops that you will identify in **Part B**.

	Name	Type	Status	Description	Delete
▶	CWATER	WATER	Input Complete		×
▶	HPSTEAM	STEAM	Input Complete	High Pressure Steam, Inlet Temp=250 C...	×
▶	LPSTEAM	STEAM	Input Complete	Low Pressure Steam, Inlet Temp=125 C,...	×
▶	MPSTEAM	STEAM	Input Complete	Medium Pressure Steam, Inlet Temp=1...	×
▶	NATGAS	GAS	Input Complete		×

Figure 2 Utilities

Based on the provided design parameters, five utilities were added: cooling water, high pressure steam, medium pressure steam, light pressure steam, and natural gas.

Part B: Identification and Design of Process Control Loops

Following the procedure that I have outlined in the lectures, calculate the Control Degrees of Freedom for this process and design all process control loops using the additional information outlined below.

Data and Assumptions

1. Both refluxes in the **DIB** column and the **Purge** column are on flow control
2. Residence time in the reflux accumulation drums: **4 min**
3. Residence time in the distillation column sumps: **2 min**
4. The waste stream is also on flow control due to environmental restrictions
5. Neither the fresh feed nor the product flow rates are fixed by other plant considerations
6. Heating value of natural gas used in the fired heater: **50.0 MJ/kg**
7. The heat capacity of water can be taken equal to **4.18 kJ/(kg C)**
8. Temperature rise in the cooling water is limited to **11 °C**

Process Variables Units for Reporting

1. Liquid flow rates must be given in **gpm**
2. Gas flow rates in **lb/hr**
3. Temperature in **deg F**
4. Pressure in **psia**
5. Height in **ft**

For each control loop you have identified write a summary similar to the template shown below:

Template: Design of Process Control Loops

Loop #	Controller Description	Controlled Variable: CV	Set Point Value: SP	Manipulated Variable: MV	Steady-State Normal Value:
1	DIB Column Reflux Flow	Reflux Flow	120.4 gpm	Reflux Flow	121.81 gpm
2	Purge Column	Reflux flow	200 gpm	Bottoms flow	200.16 gpm
3	Feed-Effluent Heat Exchanger	Temperature	118 °F	Bypass flow	118.2 °F
4	Fixed Bed Reactor	Temperature	390 °F	Feed Temperature	390 °F
5	Furnace	Temperature	280 °F	Natural Gas Flow	280 °F
6	Flooded Condenser	Pressure	650 psia	Cooling Condenser	650 psia

The total control degrees of freedom were determined to be 6. This value was calculated from the total of material and energy streams (16 streams) minus the number of extra phases subtracted by 1 (10 phases). Based on this, the control loop count was decided to be six and their values were taken directly from Aspen after being running the simulation.

Part C: Process Simulation Calculations

From pilot plant studies it was recommended to increase the reactor feed temperature by 10 °F in order to increase production rate. Change the set point of the temperature controller and answer the following questions after carrying out the simulation with the new reactor feed temperature.

1. Briefly explain your logic in setting up the simulation for the change in the temperature loop set point value from 390 deg F to 400 deg F. Did you have to change the set points of any other controllers? If yes provide a brief explanation as to why and then enter their new values in the template shown below.

The temperature was increased to 400 °F to further enhance the reaction kinetics and increase the conversion of n-butane to iso-butane. To accommodate this change, the flow rate for Stream 1 was increased by 5% to 126.4 gpm to introduce more n-butane into the system.

2. Run the simulation in order to calculate the new steady-state values of the manipulated values for all controllers and enter the new values in the template shown below

After running the simulation with the new parameters, the following data table uses the values created by Aspen. While this may not be an intended effect, it appeared much of the steady state values retained similar ones from the initial 390 °F specification. However, DIB Column Reflux Flow (Control Loop #1) saw a significant drop, decreasing from 121.81 gpm to 0.06 gpm. A similar drop also occurred for Purge Column reflux flow (Control Loop#2) which decreased to 0.15 gpm.

Template: Simulation Calculations after Set Point Change

Loop #	Controller Description	Controlled Variable: CV	Old Set Point: SP	New Set Point: SP	MV	Old Steady State Value	New Steady State Value
1	DIB Column Reflux Flow	Reflux Flow Rate	120.4 gpm	126.4 gpm	Reflux Flow Rate	121.81 gpm	0.06 gpm
2	Purge Column	Reflux flow	200 gpm	200 gpm	Bottoms flow	200.16 gpm	0.15 gpm
3	Feed-Effluent Heat Exchanger	Temperature	118 °F	118 °F	Bypass flow	118.2 °F	105. 78 °F
4	Fixed Bed Reactor	Temperature	390 °F	400 °F	Feed Temperature	390 °F	390 °F
5	Furnace	Temperature	280 °F	280 °F	Natural Gas Flow	280 °F	280 °F
6	Flooded Condenser	Pressure	650 psia	650 psia	Cooling Condenser	650 psia	650 psia